

$$6 \quad (a) \quad (i) \quad v = f\lambda \quad \Rightarrow \quad \lambda = \frac{v}{f} = \frac{320}{400} = 0.80 \text{ m} \quad [1]$$

$$(ii) \quad \text{path difference} = 7 \text{ m} - 5 \text{ m} = 2 \text{ m} = 2.5 \lambda \quad [1]$$

As there is a phase change of π during reflection, a path difference of 2.5λ would suggest a maximum at point X. [1]

$$(b) \quad (i) \quad d \sin \theta = n\lambda \Rightarrow \quad d = \frac{n\lambda}{\sin \theta} = \frac{(1)(632 \times 10^{-9})}{\frac{38}{\sqrt{(38^2 + 165^2)}}} = 2.816 \times 10^{-6} \text{ m} \quad [1]$$

$$\text{Hence, number of lines per metre} = \frac{1}{2.816 \times 10^{-6}} = 355000 \quad [1]$$

(ii) The distances PX and PY remain unchanged and P remains at the same spot. [1]

However, the positions of X & Y will be rotated by 90° . [1]

(c) The laser beam is not pointed perpendicularly to the plane of the grating. [1]

$$7 \quad (a) \quad \sin \theta = \lambda / b = (632 \times 10^{-9}) / (8.0 \times 10^{-5}) \quad [1]$$